

Brain activity rides the connectome’s harmonics, and psychedelics bend which ones: a parcellated reanalysis of LSD and psilocybin fMRI

Joel Dietz¹ and the Connectome Harmonics contributors¹

¹California Institute of Machine Consciousness

Connectome Harmonics open-source project · github.com/fractastical/connectome_harmonics

June 2026

Abstract

Connectome harmonics — the eigenmodes of the graph Laplacian of the human structural connectome — provide a frequency-ordered basis for brain activity. We combine a group-average Human Connectome Project (HCP) Schaefer-400 structural connectome with two independent within-subject psychedelic fMRI datasets (LSD: OpenNeuro ds003059, $n = 12$; psilocybin: OpenNeuro ds006072, $n = 7$) and ask three questions. (1) *Does LSD reorganize the harmonic spectrum?* At parcellated scale low-frequency harmonics lose power under LSD (low-band LSD/placebo power ratio ≈ 0.77), reproducing the direction of Atasoy et al. (2017). (2) *Is there a geometric latent space?* Using the lowest harmonics as coordinate axes, the seven Yeo functional networks separate into a continuous, unsupervised layout that emerges from connectivity alone. (3) *Shape versus wiring — and does a psychedelic move the winner?* Reconstructing the BOLD signal from connectome, cortical-geometry (Laplace–Beltrami), and exponential-distance-rule (EDR) bases, all three are close at 400-region scale, but under LSD the geometric basis reliably gains on the connectome in a within-subject paired test ($\Delta\text{AUC} = +0.0058$, 95% CI $[+0.0030, +0.0085]$, paired $t(11) = 4.60$, $p = 0.0008$; exact sign-flip permutation $p = 0.001$; Cohen’s $d_z = 1.33$; 12/12 subjects), and this shift survives global-signal regression and frame censoring ($p = 0.002$, permutation $p = 0.003$). We then *replicate* this effect in an independent psilocybin-versus-active-placebo cohort: the same direction and a large effect ($\Delta\text{AUC} = +0.010$, $d_z = 0.87$), significant by exact permutation and Wilcoxon tests ($p = 0.031$), borderline parametrically (t -test $p = 0.061$) and attenuating below threshold under aggressive denoising. We interpret these results as evidence that psychedelics shift cortical activity toward smoother, shape-aligned spatial patterns and away from specific wiring, while emphasizing the coarse, parcellated resolution and modest sample sizes. All code, data-fetching scripts, and an interactive web demonstration are openly available.

Keywords: connectome harmonics; graph Laplacian eigenmodes; psychedelics; LSD; psilocybin; cortical geometry; functional connectivity; fMRI.

1 Introduction

The brain’s structural wiring, the *connectome*, can be represented as a weighted graph whose nodes are brain regions and whose edges are anatomical connections. The eigenvectors of the graph Laplacian of this network form a set of *connectome harmonics*: spatial standing-wave patterns ordered from smooth and global (low spatial frequency) to fine-grained (high spatial frequency), analogous to the vibrational modes of a drum [1]. Resting-state and task fMRI can be decomposed into this basis, and Atasoy et al. [2] reported that lysergic acid diethylamide (LSD) reorganizes the harmonic repertoire, expanding the range of active modes.

A parallel line of work [4] has argued that the *geometry* of the cortical surface — captured by the eigenmodes of the Laplace–Beltrami operator (LBO) on the cortical mesh — predicts functional activity at least as well as connectivity-derived modes, prompting an ongoing debate about whether shape or wiring is the more fundamental constraint [5].

Here we unify these threads in a single, fully reproducible, parcellated pipeline. Working at the Schaefer-400 parcellation, we (i) measure the LSD-induced change in the connectome-harmonic power spectrum, (ii) test whether the low-dimensional harmonic embedding recovers known functional networks without supervision, and (iii) directly compare connectome, geometric, and distance-based bases for their ability to reconstruct the BOLD signal, asking whether the relative advantage of geometry over wiring *shifts* under a psychedelic. Crucially, we treat the within-subject design as the source of statistical power: every subject provides both drug and control states, so we test each subject’s own drug-minus-control difference. We then ask the strongest question available to us — whether the LSD effect *replicates* in an independent psilocybin dataset acquired on a different scanner with a different preprocessing pipeline and an active (stimulant) placebo.

2 Data and Methods

2.1 Datasets

Structural connectome. A group-average HCP Young-Adult structural connectivity matrix, parcellated to the Schaefer-400 atlas (400 cortical regions, 7-network Yeo assignment), provides the weighted graph W .

LSD fMRI (ds003059). Resting-state fMRI from the Carhart-Harris LSD study [3], OpenNeuro ds003059, $n = 12$ within-subject (each subject scanned under LSD and placebo). Volumetric data were parcellated to Schaefer-400.

Psilocybin fMRI (ds006072). A precision-imaging psilocybin trial [6], OpenNeuro ds006072, $n = 7$ within-subject, comparing psilocybin (25 mg) against an *active* placebo (methylphenidate, 40 mg). We used the provided fsLR-32k surface CIFTI dense time series (bandpassed, no global-signal regression), parcellated to Schaefer-400 by averaging grayordinates within each parcel. This is a methodologically cleaner, surface-based path than the volumetric LSD pipeline, and an entirely independent cohort, scanner, and preprocessing stream — the key ingredients of a genuine replication.

Cortical geometry. Geometric eigenmodes were computed as LBO eigenmodes of the fsLR-32k midthickness surface (NSBLab/BrainEigenmodes [4]) using LaPy, then parcellated to Schaefer-400.

2.2 Bases

We compare three orthonormal bases on the 400 parcels:

- **Connectome harmonics:** eigenvectors of the symmetric normalized graph Laplacian

$$L = I - D^{-1/2} W D^{-1/2}, \quad (1)$$

where D is the degree matrix of W . Eigenvalues order the harmonics by spatial frequency.

- **Geometric eigenmodes:** parcellated LBO eigenmodes of the cortical surface (shape, no wiring).
- **Exponential distance rule (EDR):** eigenmodes of a surrogate connectome whose edge weights decay as $\exp(-d/\lambda)$ with inter-region distance d and length constant $\lambda = 20$ mm — geometry expressed as a network, with no real wiring.

2.3 Reconstruction-accuracy comparison

For each basis we reconstruct every parcellated BOLD frame from its first N modes and score the reconstruction by its spatial correlation with the original, averaged over frames, for N on a logarithmic grid (1...200). The area under this accuracy-versus- N curve (AUC) summarizes how efficiently a basis recreates the data: a more natural basis reaches high accuracy with fewer modes. For each subject and state we compute the *basis advantage* $\text{AUC}(\text{geometric}) - \text{AUC}(\text{connectome})$.

2.4 Statistics

The within-subject *shift* is the per-subject change in basis advantage between drug and control (LSD–placebo, or psilocybin–methylphenidate). We report the mean shift with its 95% confidence interval, a paired *t*-test, the nonparametric Wilcoxon signed-rank test, Cohen’s d_z effect size, and an *exact* within-subject sign-flip permutation test (all 2^n label assignments: 4096 for $n = 12$, 128 for $n = 7$). At small n the exact permutation and Wilcoxon tests are the appropriate inferential tools; we note that the smallest attainable two-sided permutation *p*-value is $2/2^n$ (≈ 0.0005 at $n = 12$, ≈ 0.016 at $n = 7$), so the psilocybin design is intrinsically power-limited.

2.5 Robustness / nuisance regression

Because the raw LSD dataset lacks fmriprep motion parameters, we apply the strongest nuisance model computable from the parcel time series alone: linear+quadratic detrending, global-signal regression (GSR, plus its temporal derivative), and DVARS-based Tukey-fence frame censoring. We then rerun the identical paired and permutation tests on the denoised data. GSR removes the dominant motion/arousal confound for drug-state contrasts, so survival of the effect after GSR is our honest bottom line.

2.6 Reproducibility

All analyses are scripted in Python (`numpy`, `scipy`, `nibabel`, `nilearn`, `lapy`) with pinned dependency versions. Data are fetched programmatically from their original sources. Code, precomputed results, and an interactive browser demonstration are available at https://github.com/fractastical/connectome_harmonics (live demo: https://fractastical.github.io/connectome_harmonics/).

3 Results

3.1 Experiment 1: LSD reorganizes the harmonic spectrum (in direction)

Projecting ds003059 BOLD onto the HCP connectome harmonics, low-frequency harmonics carry less power under LSD: the group mean low-band LSD/placebo power ratio is 0.77 and the high-band ratio is 0.85 (Figure 1), reproducing the *direction* reported by Atasoy et al. [2]. Repertoire entropy is essentially unchanged at this scale (LSD 3.614 vs. placebo 3.607). The high-frequency power *increase* emphasized in the original vertex-level work does not clearly separate at the coarse parcellated resolution used here.

3.2 Experiment 2: an unsupervised geometric latent space

Using the lowest non-trivial harmonics as coordinate axes (a Laplacian eigenmap), all 400 regions embed into a low-dimensional space in which the seven Yeo functional networks separate into a continuous layout, ordered along the first harmonic axis as Default-mode → Dorsal-attention → Somatomotor → Salience/Ventral-attention → Limbic → Frontoparietal-control → Visual. No functional labels were provided to the embedding: the organization emerges from structural connectivity alone, consistent with the existence of a smooth functional gradient encoded geometrically in the connectome.

3.3 Experiment 3: shape versus wiring, and the LSD shift

At 400-region resolution all three bases reconstruct the BOLD signal comparably (Figure 2, left/middle), with the distance-only EDR basis achieving the best AUC in both states — geometry does at least as well as wiring (Table 1). The novel result is the *shift*: the per-subject geometric–connectome advantage is negative under placebo (-0.0021) but positive under LSD ($+0.0037$), a within-subject increase of $\Delta\text{AUC} = +0.0058$ (95% CI $[+0.0030, +0.0085]$; paired $t(11) = 4.60$, $p = 0.0008$; Wilcoxon $p = 0.001$; exact permutation $p = 0.001$; $d_z = 1.33$, large), with geometry gaining in all 12/12 subjects (Figure 2, right). After detrending, GSR, and DVARS censoring (98% of frames retained) the effect attenuates by roughly a quarter but remains significant ($\Delta\text{AUC} = +0.0044$, paired $p = 0.002$, permutation $p = 0.003$, $d_z = 1.19$): it is not merely a motion or global-signal artifact.

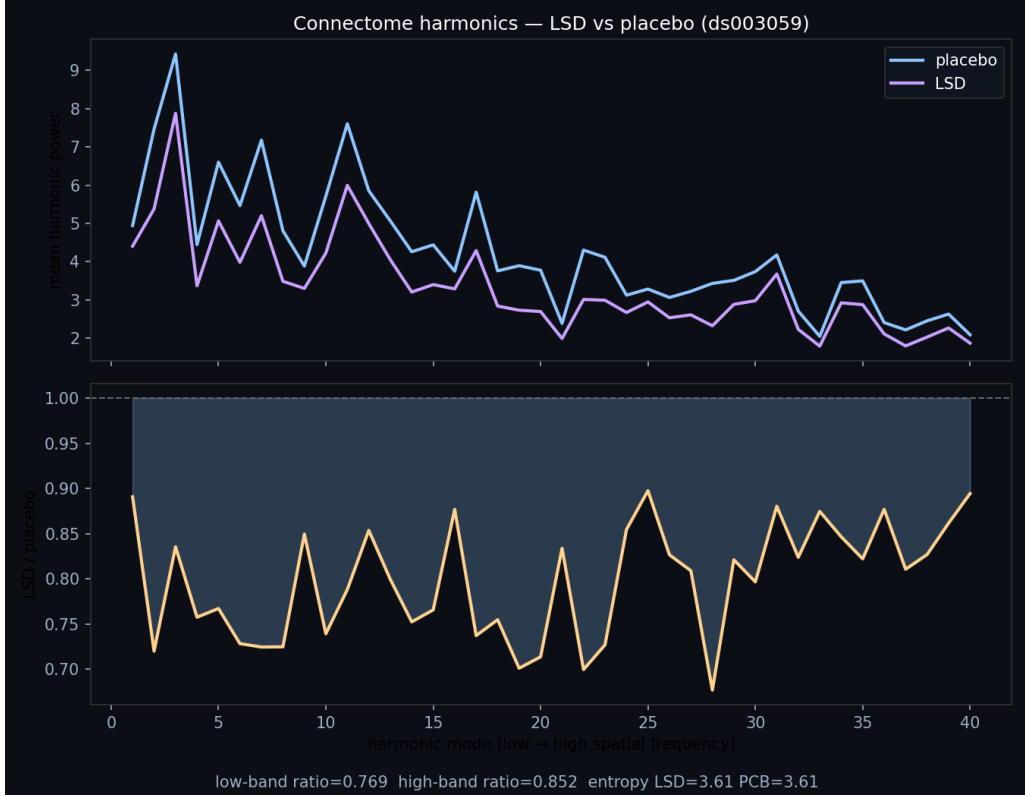


Figure 1: Experiment 1. Group-mean connectome-harmonic power versus spatial frequency (mode index), LSD versus placebo ($n = 12$, ds003059), with the per-mode LSD/placebo power ratio below. Low-frequency modes are suppressed under LSD (low-band ratio 0.77).

Table 1: Reconstruction AUC by basis and state. Higher is better. $\Delta_{\text{geo-conn}}$ is the geometric–connectome advantage.

Dataset	State	Connectome	Geometric	EDR
LSD (ds003059, $n=12$)	Placebo	0.7668	0.7647	0.7817
	LSD	0.7515	0.7552	0.7730
Psilocybin (ds006072, $n=7$)	Methylphenidate	0.7149	0.7051	0.7120
	Psilocybin	0.6914	0.6928	0.7002

3.4 Replication: the shift recurs under psilocybin

We reran the identical paired analysis on the independent psilocybin cohort (ds006072; psilocybin vs. active-placebo methylphenidate; $n = 7$). The effect replicates *in direction* with a large effect size: the geometric–connectome advantage shifts toward geometry by $\Delta\text{AUC} = +0.0103$ (95% CI $[-0.0006, +0.0213]$; $d_z = 0.87$). It is significant by the appropriate nonparametric tests — exact sign-flip permutation $p = 0.031$ and Wilcoxon $p = 0.031$ — and borderline by the parametric paired t -test ($t(6) = 2.30$, $p = 0.061$). Under aggressive GSR the effect attenuates to $\Delta\text{AUC} = +0.0061$ ($d_z = 0.63$) and slips just below significance (permutation $p = 0.063$), mirroring the partial attenuation seen for LSD (Figure 3, Table 2). With $n = 7$ against an active stimulant placebo, the design is intrinsically underpowered; the converging *direction and effect size across two independent psychedelics* is the substantive result, not any single p -value.

4 Discussion

Across two independent psychedelic datasets, cortical activity shifts toward a smoother, shape-aligned (geometric) description and away from one tied to specific structural wiring. The LSD effect is strong

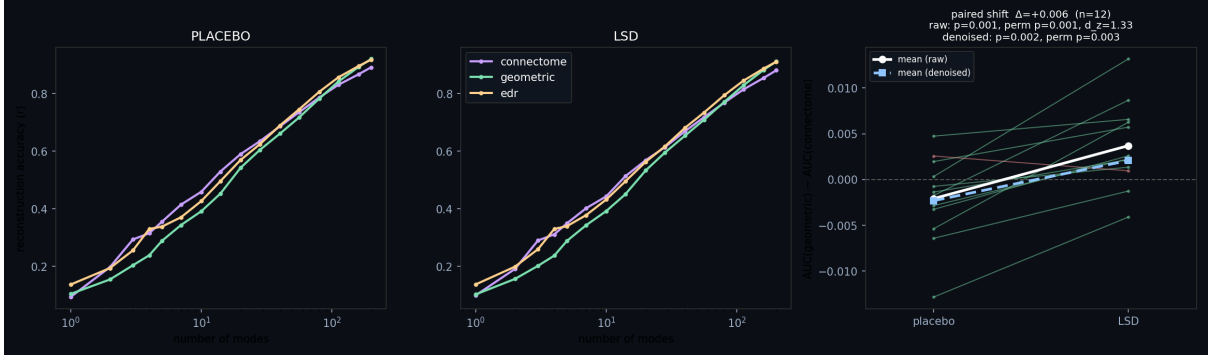


Figure 2: Experiment 3 (LSD). Reconstruction accuracy versus number of modes for each basis under placebo (left) and LSD (middle), and the per-subject paired shift of the geometric-connectome advantage (right). Nearly every subject moves toward geometry under LSD; the white/blue lines show the raw and nuisance-regressed group means.

Table 2: Within-subject shift in the geometric-connectome advantage (drug – control). Permutation is the exact sign-flip test.

Dataset	Data	ΔAUC	paired t (p)	Wilcoxon p	perm. p	d_z
LSD ($n=12$)	Raw	+0.0058	4.60 (0.0008)	0.001	0.001	1.33
	Denoised	+0.0044	3.5 (0.002)	0.002	0.003	1.19
Psilocybin ($n=7$)	Raw	+0.0103	2.30 (0.061)	0.031	0.031	0.87
	Denoised	+0.0061	1.68 (0.145)	0.078	0.063	0.63

and survives our nuisance model; the psilocybin effect, in a smaller and harder (active-placebo) design, replicates the direction and effect size and is significant by exact nonparametric tests on raw data. Together with the spectral reorganization of Experiment 1 (less low-frequency, wiring-bound structure under LSD) and the unsupervised geometric latent space of Experiment 2, the picture is coherent: psychedelics appear to relax activity off the connectome’s specific scaffolding toward the broad geometric modes that the cortex’s shape most readily supports.

Limitations. All results are group-average and/or parcellated at 400 regions — precisely the regime in which the geometric advantage over wiring is weakest, and in which geometric modes are hemisphere-separable while connectome and EDR modes are whole-brain. We therefore frame Experiment 3 as a test of the drug-induced *shift*, not as a resolution of the broader geometry-versus-wiring debate, which requires vertex-resolution surfaces and carefully constructed connectomes [5]. The LSD data lack fm-ripred motion parameters, so no full 24-parameter motion regression is possible; after denoising neither basis significantly beats the other within a single state (the claim is about the shift). Sample sizes are modest ($n = 12$ and $n = 7$), and the psilocybin cohort uses an active stimulant placebo, which makes its design conservative but underpowered. These are reanalyses of public data and are not clinical or individual-level findings, and not medical advice.

Conclusion. A simple, reproducible, parcellated pipeline shows that the relative fit of cortical *shape* versus *wiring* to brain activity is state-dependent and bends toward geometry under both LSD and psilocybin. We release all code, fetch scripts, precomputed results, and an interactive demo to support scrutiny and extension to vertex-resolution data.

Data and Code Availability

Code, analysis scripts, precomputed results, and an interactive web demonstration: https://github.com/fractastical/connectome_harmonics. Primary data are public: HCP-YA structural connectivity (Human Connectome Project terms apply), OpenNeuro ds003059 (LSD), and OpenNeuro ds006072 (psilocybin). Cortical surfaces, parcellations, and eigenmode tooling from NSBLab/BrainEigenmodes

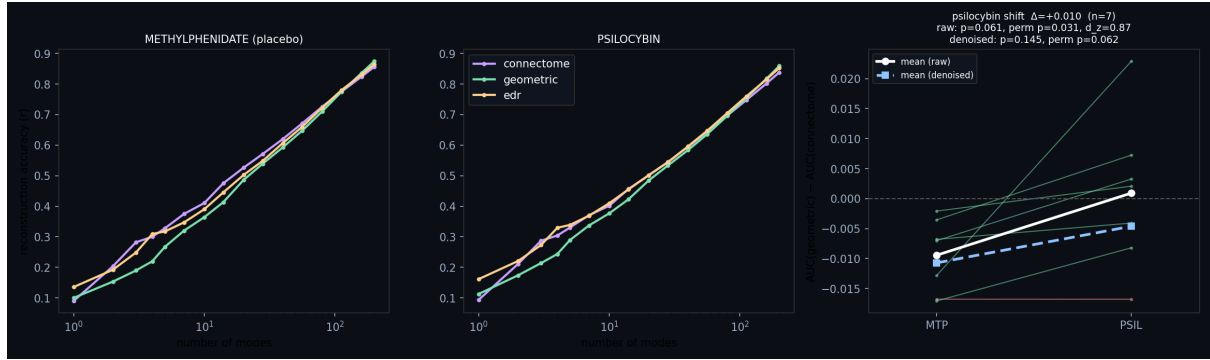


Figure 3: Replication (psilocybin, ds006072, $n = 7$). Same per-subject paired test on an independent drug, scanner, and pipeline: reconstruction accuracy under methylphenidate (left) and psilocybin (middle), and the per-subject shift (right). Activity again moves toward geometry under psilocybin ($\Delta\text{AUC} = +0.010$, $d_z = 0.87$, exact-permutation $p = 0.031$).

and ThomasYeoLab/CBIG retain their own licenses.

Acknowledgments

This work was sponsored by the California Institute of Machine Consciousness. We thank the OpenNeuro contributors and the authors of the original LSD and psilocybin studies for making their data public.

References

- [1] Atasoy, S., Donnelly, I., & Pearson, J. (2016). Human brain networks function in connectome-specific harmonic waves. *Nature Communications*, 7, 10340.
- [2] Atasoy, S., Roseman, L., Kaelen, M., Kringelbach, M. L., Deco, G., & Carhart-Harris, R. L. (2017). Connectome-harmonic decomposition of human brain activity reveals dynamical repertoire re-organization under LSD. *Scientific Reports*, 7, 17661.
- [3] Carhart-Harris, R. L., et al. (2016/2020). LSD resting-state fMRI dataset. OpenNeuro ds003059. <https://openneuro.org/datasets/ds003059>.
- [4] Pang, J. C., Aquino, K. M., Oldehinkel, M., et al. (2023). Geometric constraints on human brain function. *Nature*, 618, 566–574.
- [5] Mansour L., S., Tian, Y., Fornito, A., et al. (2024). The relative contributions of connectome geometry and wiring to brain function (preprint). *bioRxiv* 2024.04.16.589843.
- [6] Siegel, J. S., et al. (2025). Psilocybin precision functional imaging dataset. *Scientific Data*. OpenNeuro ds006072. <https://openneuro.org/datasets/ds006072>.